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IS HEAT STRESS PROCEDURE

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1.0 PURPOSE

The purpose of this procedure is to protect workers from potential adverse effects of overexposure to heat and to prevent heat related illness. Operations involving high air temperatures, radiant heat sources, high humidity, direct physical contact with hot objects, or strenuous physical activities have a high potential for inducing heat stress in employees engaged in such operations. This heat stress procedure is written to assess, inform, provide awareness, document and evaluate heat stress in the workplace.

2.0 SCOPE AND APPLICABILITY

This procedure applies to all QatarEnergy-IS Employees and Contractors

3.0 ROLES AND RESPONSIBILITIES

SITE PERSON IN CHARGE (OIS, WHM Supervisor etc.)/ SECTION HEAD

- Ensure that the Heat Stress Procedure is communicated and adhered to whilst employees are working in heat stress conditions.
- Ensure the heat stress indicators are strategically placed, updated and maintained to communicate the relevant heat index for employees and Contractors.
- Coordinate heat stress audits for the facility and ensure they are completed as per the Plan.
- Review all heat stress audits and sign off to confirm his recognition of highlighted issues or pending action.
- Provide guidance to QatarEnergy-IS and Contractors site personnel.
- Assess all activities that are required to be carried out where the heat index is measured at $\geq 54^{\circ}\text{C}$ and determine if it meets the definition of "Critical Activities" and approve or reject accordingly.
- Refer to Appendix H for Heat Stress Index Alert responsibilities.

OCCUPATIONAL HEALTH & PARAMEDIC

- Prepare information relating to heat stress and develop and be responsible for the heat stress procedure.
- Address heat stress concerns of employees and assist in provision of information sessions.
- Ensure calibrated HSE heat stress monitors are distributed to all QatarEnergy-IS locations prior to the heat stress campaign.

SAFETY OFFICER

- Ensure employee and Contractor heat stress awareness through presentation at Stand Downs for Safety, HSE meetings or performing Awareness throughout the heat stress campaign.
- Distribute leaflets and posters during Heat Stress Awareness Campaigns.

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- Assist in heat stress audits at worksite locations throughout the heat stress campaign.
- Ensure heat stress audits are signed off by the responsible person, loaded in the database and actions tracked to completion.
- Refer to Appendix H for Heat Stress Index Alert Responsibilities.
- Issue heat stress Alert as per Appendix H through Email and PAGA system. Additional information / actions may be added in the Alert, as required.

HSE MANAGER

- Announce the commencement and conclusion of the heat stress campaign to all QatarEnergy-IS facilities.

IS EMPLOYEES AND CONTRACTORS

- Be familiar with heat stress hazards, follow safe working procedures and drink enough fluids to replace those lost through perspiration.
- Follow recommended schedule of rest breaks, as advised by supervisors, to avoid heat exhaustion and collapse.
- Report to their supervisor and Offshore Medic, heat stress-related symptoms in themselves or their co-workers and follow recommended rest/ break schedule.
- Contractor personnel performing work on QatarEnergy-IS facilities shall have the necessary heat stress awareness. They shall follow the guidance provided by QatarEnergy-IS person in charge and ensure exposure to heat stress is minimize.
- Submit the Medications declaration to Offshore medic upon arrival.
- Comply with the Heat Stress Index Alert requirements and actions.

OFFSHORE DEPARTMENT LEAD / SUPERVISORS / PTW CONTROLLER

- Refer to Appendix H: Heat Stress Index Alert Responsibility
- Ensure compliance with the Heat Stress Index Alert (Appendix I) requirements and actions.
- Reassess work activity requirements and provisionally suspend the relevant permits to work in consideration of heat index ranges as required.

4.0 PROCEDURAL STEPS

4.1 FACTORS EFFECTING HEAT STRESS

Individual Susceptibility

The environment factors resulting in heat illness include more than ambient air temperature. Radiant heat air movement, conduction, and relative humidity all affect and individuals response to heat. Age, weight, degree of physical fitness and acclimatization, metabolism, use of alcohol and drugs, and a variety of medical conditions such as hypertension all affect a person's sensitivity to heat. Even the type of clothing worn shall be considered. Prior heat injuries predispose an individual to additional injury.



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Temperature

The higher the air temperature, the less heat the body can lose by convection, conduction and radiation (see definitions).

Air Humidity

The body's natural cooling system is through evaporative cooling of sweat. As the amount of moisture present in the air increases, the evaporation rate of sweat decreases and the body's cooling system becomes less efficient.

At very high humidity levels, evaporative cooling no longer takes place, and the body cannot naturally cool itself by this process.

Wind speed

When the air temperature is less than the body temperature 37 °C air movement can help personnel stay cooler by increasing both the convective heat exchange (the exchange between the skin surface and the cooler surrounding air) and the rate of evaporation.

However, at temperatures above 37 °C, evaporative cooling may be offset by addition of convective heat from the hot air. In conditions of high temperature and high humidity, air movements will in fact make the worker hotter.

Medications

Certain medications could potentially increase of Heat Stress and hence declaration of medications by the individuals shall be submitted to the Offshore medic upon arrival.

4.2 HEALTH EFFECTS AND SYMPTOMS OF HEAT STRESS

Sunburn

Sunburn is often overlooked as a danger when working outdoors in direct sunlight. Besides the discomfort of the burn itself, sunburn can prevent your body from eliminating heat efficiently and can contribute to one of the more dangerous heat disorders.

Heat rash

Heat Rash, also known as prickly heat, is another minor condition that can lower the body's ability to lose heat.

Heat cramps

Heat Cramps are a danger signal since they may occur alone or be combined with one of the other major heat distress orders. These are painful, sometimes severe, cramps of the muscles used while working, such as the arms, legs, or stomach. They often don't occur until relaxing after work. These cramps may be connected to heat, dehydration, and poor conditioning, rather than to lack of salt or other mineral imbalances.

Heat exhaustion

Heat Exhaustion occurs when the body's heat-control mechanism is over active but hasn't broken down completely. The signs of heat exhaustion include paleness,



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dizziness, nausea, vomiting, fainting, and a moderately increased temperature. The victim may also be having heat cramps, and there is a high risk that the victim will continue on to a state of heat stroke. This disorder also causes special risk to older employees or those with coronary artery disease or emphysema. Rest and water may help in mild heat exhaustion, and ice packs and a cool environment may also help.

More severely exhausted people may need IV fluids, especially if vomiting keeps them from drinking enough.

Note: Air conditioning or a cool environment is the number one protective factor against heat exhaustion.

Heat Stroke

Heat Stroke is the most severe form of heat related illnesses and is an emergency requiring immediate medical attention. Outwardly, it may first progress through the symptoms of heat cramps and/or heat exhaustion, with a dramatically sudden onset of heat stroke symptoms followed by rapid deterioration of the victim.

It can occur even in people who are not performing physical exertion, if the weather is hot enough. These people have warm, flushed skin, do not sweat, may be delirious, unconscious, or having seizures.

If a person has these symptoms, call for immediate medical attention.

Heat Syncope

Fainting, dizziness, or light-headedness, after prolonged standing or suddenly rising from a sitting or lying position.

4.3 HEAT STRESS MONITORING

The environmental factors specifically taken into account for the monitoring of heat stress are dry air temperature and relative humidity. Both parameters are combined on the heat index chart (Appendix D).

The program for controlling heat-related illness requires colored heat stress indicators (i.e. flags, boards or cards) to be displayed at strategic locations throughout the worksite. The indicator denotes the prevailing heat stress condition. Heat stress conditions are monitored throughout the day and the heat stress indicators changed as required.

The QatarEnergy-IS location person in charge is responsible for ensuring actual air temperature and humidity data is periodically monitored and reviewed and the heat stress index communicated to employees and Contractors via the heat stress indicators. Supervisors will ensure that the appropriate rest levels are allowed dependent on the heat stress indicator, and that sufficient shelter and fluids are available (Appendix E - Offshore Heat Stress Control Matrix).

Onshore locations (e.g. Ras Laffan) shall refer to Onshore Heat Stress Control Matrix (Appendix G).

The Heat Stress campaign will commence when recorded Heat Index is within the Orange zone. The start and the conclusion of the campaign will be announced by the HSE Manager and communicated to all QatarEnergy-IS locations.



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The performing party should check Heat Index at the specific work location, since wind and other factors may contribute to Heat Index level

4.4 PREVENTION AND CONTROL OF HEAT STRESS

General Ventilation - Ventilation, air cooling, fans, shielding and insulation are the five major types of engineering controls used to reduce heat stress in hot working environments. Outdoor work areas need to have a shaded area accessible to employees.

Fluid Replacement - Cool (10-15°C) water or any cool liquids (non-alcoholic) shall be made available to workers (at a minimum 250 ml every twenty minutes). Isotonic drinks and electrolyte replacements are not recommended due to the effect of increasing a person's urine output.

Personal Protective Equipment – Cool neckerchiefs/ bandanas, hardhat cool pad inserts, dark safety glasses and sunscreen shall be worn during the heat stress campaign.

Work/Rest Cycles – Heavy and minimal work activities shall be alternated. Tasks shall be rotated amongst employees (Appendix E, F and G).

Heat Stress Audits – Heat Stress Audits (Appendix C) are completed to monitor environmental temperature, humidity levels and the site management of heat stress campaign at QatarEnergy-IS locations during the heat stress campaign. The audits shall be loaded into the Audit/Inspection Action Tracker database on the HSE team site for action tracking purposes and analysis. The QatarEnergy-IS person in charge of the work site is responsible to ensure a minimum of three heat stress audits are completed weekly during the heat stress campaign at their location. The HSE Advisor shall be responsible for that location, ensuring audits are signed off by the QatarEnergy-IS Representative, loaded into the database and actions tracked to completion.

Heat Stress Equipment – Equipment such as a Heat/humidity meter shall be made available at all QatarEnergy-IS work sites for measuring temperature and relevant humidity during heat stress campaign. The heat stress equipment shall be calibrated and distributed by the Occupational Health & Paramedic prior to the commencement of the heat stress campaign.

Incident Reporting – Potential heat stress incidents shall be communicated immediately to the QatarEnergy-IS via Occupational Health & Paramedic or Safety Officer. Incident reports shall be documented in accordance with QatarEnergy Procedure for HSE Incident Reporting Investigation Learning (QP-HSE-PRC-022).



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4.5 TREATMENT FOR HEAT STRESS

Heat cramp

- Stop work immediately and move the person to a cool shady resting place.
- Give patient enough cool drinking water, juice or oral rehydration solution and let patient rest for at least 8 hours.

Heat exhaustion

- Move person to a cool shady resting place - do not leave the patient alone.
- Cool the person rapidly with running water, cold compress or rapid fanning; Remove extra clothing when necessary.
- Give cool drinking water or oral re-hydration solution. Avoid caffeine or alcohol and let the patient rest for at least 12 hours.

Heat stroke

- Immediately call for emergency medical assistance/treatment.
- Place the person in a cool shady area and do not leave them alone
- Cool the patient rapidly with running water, cold compresses or rapid fanning Remove the patient's outer clothing.
- Provide cool drinking water if the patient is alert - Do not give caffeine or alcohol.
- Continue to cool body temperature until medical assistance arrives and patient can be taken to a medical facility for further cooling and monitoring of body temperature.

4.6 HEAT STRESS AWARENESS

The following heat stress awareness topics shall be discussed with employees and Contractors during the heat stress campaign months and detailed during health campaigns via pre-job toolbox talks, QatarEnergy-IS safety stand downs, HSE Meetings and Awareness training.

- Acclimatization and hydration awareness.
- Knowledge of the hazards of heat stress.
- Recognition of predisposing factors, danger signs, symptoms and potential effects of heat stress related illnesses and injuries.
- Awareness of first-aid procedures for treatment of heat stress related illnesses and injuries including heat stroke and heat cramp.
- Employee responsibilities in avoiding heat stress.
- Dangers of using drugs including therapeutic ones, and alcohol in hot work environments.
- Use of protective clothing and equipment.
- Planning of strenuous physical activities and hot operations during cooler periods, wherever possible (e.g. early morning or evening/ night);
- Importance of regular rest breaks, in cool shade or air-conditioned during hot periods
- Requirement to drink fluids regularly to replace body fluids lost through sweating.

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4.7 ACCLIMATIZATION

Acclimatization is a gradual physiological adaptation that improves an employee's ability to tolerate heat stress. The process requires several days with gradual build-up of tolerance. QatarEnergy and Contractors shall adhere to acclimatization process as per (Appendix J)

4.8 HEAT STRESS ALERT

- The Matrix provided in Appendix E shall be followed by the concerned personnel
- Heat Stress Index Alert Template provided in Appendix I shall be followed. Additional information / actions may be added in the Alert, as required

5.0 REFERENCES TO OTHER DOCUMENTS

Document Title	Document Number	Internal/External Document (if any)	Document Referencing
Standard for Managing Heat Stress	QP-OHH-STD-012	Internal	Upward
HSE Incident Reporting Investigation Learning	QP-HSE-PRC-022	Internal	Sideward

6.0 DEFINITIONS

Term	Definition
Conduction	Transfer of heat through contact. Heat passes from the warmer material to the cooler material. For example, a person skin can transfer heat to a contacting surface if that surface is cooler, and vice versa.
Convection	Transfer of heat in a moving fluid. Air flowing past the body can cool the body if the air temperature is cooler. On the other hand, air that exceeds 37°C can increase the heat load of the body, especially when humidity is high.
Radiation	Transfer of heat energy through space. A person whose body temperature is greater than the temperature of the surrounding surfaces radiates heat to these surfaces. A person will also feel radiation heat from the sun when exposed to direct sunlight.
Critical Activities	As a general guide, Business Critical Activities which are required in order to maintain the continuity of Oil & Gas Operations & HSE Critical Activities which include construction, installation, commissioning, operation, modification, repair, inspection, testing or examination activities associated with assuring the integrity of HSE Critical Equipment/ Systems.
Elevated Work	As a general guide, A working location with an elevation of six (6) feet or 1.8 meters above the floor or other working surface and requires the use of personal fall protection, rope access or climbing of ladders.
Heat Stress	Heat stress occurs when heat absorbed from the environment is faster than

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Term	Definition
	the body can get rid of it. The resulting strain on the body comes from the combined contributions of the job (e.g., work activity), environmental factors (e.g., air temperature, humidity, air movement, radiant heat), and worker factors (e.g., extent of acclimatization and hydration)
Heat Index	An index that combines air temperature and relative humidity in an attempt to determine the human-perceived equivalent temperature

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APPENDIX**APPENDIX A - REVISION HISTORY LOG****Revision Number: 01****Date: 19/03/2023**

Item Revised	Revision Description	Page No.
Common	Updated based on new Management System Documentation Format and current organization structure	All
3.0 Roles and Responsibilities	Included additional responsibility under SITE PERSON IN CHARGE (OIS, WHM Supervisor etc.)/ SECTION HEAD pertaining to work activities during heat index of 54°C and above	3
6.0 Definitions	Included definitions for Critical Activities and Elevated Work	8

Revision Number: 02**Date: 12/07/2023**

Item Revised	Revision Description	Page No.
4.2	Added definition of Heat Syncope	4
Appendix G, H and I	New Appendix added on Heat stress Alerts and Heat Acclimatization	15-19

Revision Number: 03**Date: 14/08/2023**

Item Revised	Revision Description	Page No.
Appendix E	Updated to align with Corporate HSE Standard (QP-OHH-STD-012)	4

Revision Number: 04**Date: 01/10/2023**

Item Revised	Revision Description	Page No.
Appendix E	Aligned with Corporate Standard For Managing Heat Stress (QP-OHH-STD-012).	14
Appendix E	Added IRATA requirements for Thermal work limit	15

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APPENDIX B – LIST OF ABBREVIATIONS

Abbreviation	Description
QatarEnergy-IS	Idd El-Shargi Asset of Qatar Energy
OIS	Offshore Installation Superintendent
WHM	Wellhead Maintenance
Shall	Mandatory action
Should	Recommended action
TWL	Thermal Work Limit

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APPENDIX C: HEAT STRESS AUDIT

Location:

Temperature				Y/N	Comments
Record air temperature, humidity & index rating:					
Any additional heat source that may add to the heat loading					
Is there a heat source on site that may add to the heat loading?					
Are extra precautions required because of additional heat source?					
Worksite				Y/N	Comments
Is the work site in a shaded area?					
Has shade protection been provided? Is it required?					
Is the work being carried out strenuous? Detail the work activity, Location and Permit No# if applicable;					
Light	i.e. Cleaning	Medium	i.e. Hammering	Heavy	i.e. Manual handling
Has the heat index color code been displayed and updated?					
Aware of color code rest period requirements and breaks applied?					
Was work postponed until a cooler part of the day?					
Drinking water available on site or nearby?					
Personnel aware of heat stress precautions and can detail signs?					
Buddy/rotation system in use for activity and monitoring each other?					
Aware of color code water intake requirements?					
Have mechanical aids been provided for manual handling?					
Are personnel using cooling aids?					
First Aid				Y/N	Comments
Trained first aiders available?					
Personnel aware of the nearest medical facility?					
Are urine charts posted in the toilets – are they multi-lingual?					
Sample Checks – Pick an individual and perform a review on the following:					
Individuals body temperature using the non-contact thermometer				Temp:	
Carry out the pinch test on the back of the hand					
Check if the person is thirsty and advise					
Check how much water the individual has consumed that day.				Volume:	
Advise the individual on any finding that may lead to a heat stress issue					
Auditor		Signature		Date	
Site In charge		Signature		Date	



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APPENDIX D: HEAT STRESS INDEX

	RELATIVE HUMIDITY (%)									
		10%	20%	30%	40%	50%	60%	70%	80%	90%
AIR TEMPERATURE (°C)	50	44	52	54	54	54	54	54	54	54
	49	43	51	54	54	54	54	54	54	54
	48	43	50	53	54	54	54	54	54	54
	47	42	48	52	54	54	54	54	54	54
	46	41	47	50	54	54	54	54	54	54
	45	41	46	50	54	54	54	54	54	54
	44	40	45	49	54	54	54	54	54	54
	43	39	44	48	54	54	54	54	54	54
	42	38	43	46	54	54	54	54	54	54
	41	38	41	45	52	54	54	54	54	54
	40	37	40	43	49	54	54	54	54	54
	39	36	38	42	47	52	54	54	54	54
	38	35	37	41	43	49	54	54	54	54
	37	34	36	38	41	43	54	54	54	54
	36	33	35	37	40	42	49	53	54	54
	35	32	34	36	39	42	46	52	54	54
	34	32	34	35	38	41	44	50	53	54
	33	31	32	33	36	38	41	46	50	53
	32	29	31	32	33	36	38	41	46	50
	31	28	30	31	32	34	35	38	41	46
	30	27	29	29	31	32	33	36	38	43
	29	26	27	28	29	30	32	33	35	37
	28	25	26	27	27	29	30	32	33	35
	27	24	25	26	26	27	28	29	30	32
	26	22	24	25	26	26	27	27	28	29

SEVERITY	HEAT INDEX	HEAT SYNDROME
IV. Extreme Danger	> 54	Heat Stroke or Sunstroke imminent
III. Danger	39 - 53	Sunstroke, Heat Cramps or Heat Exhaustion likely, Heat Stroke Possible with Prolonged exposure and physical activity
II. Extreme Caution	32 - 38	Sunstroke, Heat Cramps or Heat Exhaustion possible with Prolonged exposure and physical activity
I. Caution	22 - 31	Fatigue possible with Prolonged exposure and physical activity

Note:

1. Degree of Heat Stress may vary with age, health and body characteristics.
2. Do not take any form of "Salt Supplement" unless advised.

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APPENDIX E: OFFSHORE HEAT STRESS CONTROL MATRIX

Heat Index (Deg C)	Work: Rest (Minutes)	Water Requirements	Controls	Flag Colour
54 or higher*			STOP all work, Work requirements evaluation (Note*)	Red (Extreme Danger)
50:53 (Upper Orange)	20:10	250 ml every 10 mins	STOP all Elevated work	Orange (Danger)
39:49 (Lower Orange)	30:10	250 ml every 15 mins	Work under shade No Confined space work (Note**)	
32-38	40:10	250 ml every 20 mins	No working alone	Yellow (Extreme Caution)
27-31	50:10	250 ml every 20 mins	Continuous visual monitoring of workers under direct sun and heavy work	Green (Caution)
Note*	Critical activities will be allowed to continue subject to a maximum of 20:20 work: rest periods and an approved risk assessment and additional control measures in place (i.e., air blowers, buddy system, etc.) Non-critical activities will be stopped.			
Note**	<u>Heat Index from 39 to 49 Deg C (Lower Orange)</u> - The following Elevated work will be permitted under shade with additional controls such as Personal cooling aid, workers rotation and approved Heat stress risk assessment: - Scaffold operations (Erection, dismantling and modifications) and working on Scaffold platforms (Green tag). - Rope access activities with hourly monitoring of Heat Index and TWL (Thermal Work limit) at the site location. These parameters shall be measured and monitored by Contractor Supervisor or Contractor Safety Officer. - No elevated work is permitted under the direct Sun. - No Confined space works. <u>Heat Index from 50 to 53 Deg C (Upper Orange)</u> All Elevated work shall be stopped			
Note (For RLIC)	For Orange Zone Heat Index (39-53): RLIC pipe yards where working under shade is not possible as a heat stress control shall be individually risk assessed by the Site In-charge and additional control measures such as increased rest periods implemented.			

Additional requirements

- Rest intervals are for rest in air-conditioned/shaded areas. Shaded break areas shall be kept cooler than the work areas by at least 10°C to 15°C. Where cooling is not possible, air circulation with fans shall be used to facilitate cooling by sweat evaporation (Ref. clause 3.2.1.2 of QatarEnergy Standard for Managing Heat Stress).

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2. The Rope access contractors shall comply with TWL (Thermal Work limit) requirements as per IRATA ICOP (International code of practice for industrial rope access).
Part 3: Informative annexes; Annex O: Protecting rope access technicians against environmental conditions. Summary provided in Annexure- H.
In case of conflict of requirements between QatarEnergy Heat Stress Control Matrix and IRATA, the more stringent requirement shall be applicable.
3. Rope access Contractor Supervisor shall submit the Heat Index and TWL readings to QatarEnergy site Safety officer everyday for review and records.
4. Rope access Contractor shall provide TWL meters to their personnel at site.
5. Each heat index category has specific controls that continues over onto higher indexes, meaning current heat index controls in addition to lower heat index controls shall be complied. Example: While working in upper orange zone, the Personal shall comply with Heat Index 50:53 controls in addition to Heat Index 27:49 controls.

APPENDIX F : THERMAL WORK LIMIT FOR ROPE ACCESS TECHNICIANS

Table O.4.3 – Thermal work limit (TWL) and working zones for rope access technicians

Working zones	Interventions (NB No rope access technician to work alone)	Rehydration schedule (per hour)	Work/rest schedule (minutes)
Low risk unrestricted zone TWL: 140 to <220	No limits on self-paced work	Light work: 600 ml to 1 litre	Safe for all continuous self-paced work
Medium risk cautionary zone TWL: 115 to 140	Cautionary zone indicates situations in which environmental conditions require additional precautions. - Implement practicable engineering control measures to reduce heat stress, e.g. provide shade; improve ventilation. - Ensure adequate fluid intakes appropriate to the type of work.	Light work: 1 to 1.2 litres	Safe for continuous self-paced light work
		Heavy work: >1.2 litres	Continuous paced work: 45 work/15 rest
High risk zone TWL: <115	Strict work/rest cycle required - No unacclimatized person to work. - High risk induction required emphasising hydration and identifying signs of heat strain. - Provide personal water bottle (2 litre capacity) to be on-site at all times.	All work: >1.2 litres	Light work: 45 work/15 rest
			Heavy work: 20 work/40 rest

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APPENDIX G: ONSHORE HEAT STRESS CONTROL MATRIX

Heat Index (Combination of air temperature and relative humidity)	Work : Rest Period (minutes)	Water Requirements (1 cup= 250 ml)	Controls	Flag colour
27-31	50:10	1 Cup every 20 minutes	Continuous Visual Monitoring of workers under Direct Sun and Heavy work	Green (Caution)
32-38	40:10	1 Cup every 20 minutes	No working alone	Yellow (Extreme Caution)
39-49	30:10	1 Cup every 15 minutes	Work under shade	Orange (Danger)
50-53	20:10	1 Cup every 10 minutes	Stop elevated and confined space work.	
At 54 and above	---	----	Stop all work.	Red (Extreme Danger)

Note: Rope access requirement mentioned in Appendix - E shall be complied , as applicable

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APPENDIX H: HEAT STRESS INDEX ALERT RESPONSIBILITIES

No	Task	Responsibility	Facility Safety Officer	Offshore Department Lead/ Supervisors	PTW Controller	OIS	Onshore HSE
1	Conduct daily temperature reading during heat stress period as per IS-HSE-PRC-031 (Times)		R				
2	Complete heat index alert as per heat index		R				
3	Email Heat Index Alert as per distribution list		R	I	I	I	I
4	Make PS1 Platform PA Announcement (Orange/Red)		R	I	I	I	I
5	Implement Heat Index Alert Controls		R	R	R	I	I
	Heat Index Alert Orange						
6	Confined Space Permits Suspended		R	R	R	I	I
7	Conduct risk Assessment based on critical activities and further controls to reduce risk		R	C	C	A	I
	Heat Index Alert Red						
8	Stop all work activities (Permit Suspended)		R	R	R	A	I
9	Conduct risk Assessment based on critical activities and further controls to reduce risk		R	C	C	A	I
			Abbreviations				
			R	Responsible			
			A	Approver			
			C	Contributor			
			I	Information Only			

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APPENDIX I: HEAT STRESS INDEX ALERT TEMPLATE

(A) ORANGE ZONE

Heat Stress Alert (Location)	
Current Heat Stress Index is (Sun) Deg C	45
Color	Orange (Danger)
Temperature Reading At	Hrs
Immediate Action	
Stop Confined Space work	
1	All Activities shall be conducted under the SHADE
2	Implement 30 mins work – 10 min Rest upto Heat Index of 49 Deg C (Lower Orange)
3	Implement 20 mins work – 10 min Rest for heat index from 50 to 53 Deg C (Upper Orange)
4	Reassess the Elevated Work and put additional controls
5	Water Consumption 250 ML (1 Cup) every 10 Mins
6	Heat Stress Items to be used (Cooling Aids, Fans)
7	Stop all elevated work once the Heat Index touches 50 Deg C
Heat Stress Symptoms	
A	Heat Cramps
B	Heat exhaustion
C	Heat Stroke
For further guidance on Heat Stress controls please contact any of the HSE Personnel	
Department Supervisors are responsible for sharing this alert with their team and implementing the correct controls	

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(B) RED ZONE

PS1 Heat Stress Alert	
Current Heat Stress Index is (Sun) Deg C	54
Colour	RED (Extreme Danger)
Temperature Reading At	hrs
Immediate Action	
STOP ALL WORK	
Work Requirements Evaluation	
EXTREME HEAT WARNING	
Implement	
1	Non-Critical Work Activities will be "Stopped"
2	Critical Work Activities (personal, plant safety and high production loss may be authorized once an approved risk assessment and additional control measure is in place) - Subject to OIS Approval & PS1 Safety Officer review and only for a period of 20 min work - 20 min rest.
Note: These are in additional controls to be implemented in addition to the Controls required for Orange Zone.	
For further guidance on Heat Stress Prevention controls please contact any of the HSE Personnel	
Department Supervisors are responsible for sharing this alert with their team and implementing the correct controls	

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APPENDIX J - ACCLIMATIZATION PROCESS FOR WORK PRODUCTIVITY

Reference: QP-OHH-STD-012 (Standard For Managing Heat Stress)

<u>Acclimatization</u>					
Day	Activity (as percentage of full work assignment)				
	Experienced worker	New Worker			
1	50%	20%			
2	60%	40%			
3	80%	60%			
4	100%	80%			
5	100%	100%			
<u>Re-acclimatization</u>					
Absence	Illness (days away from heat related job routine)	Exposure sequence (percent of full work assignment per day)			
		Day 1	Day 2	Day 3	Day 4
<4	-	100%			
4-5	1-3	*R/E	100%		
6-12	4-5	80%	100%		
12-20	6-8	60%	80%	100%	
>20	>8	50%	60%	80%	100%
*R/E: Reduced expectations by workers and supervision					

**Source: this table is adopted from OGP IPIECA Report No. 398*